

$$\begin{aligned}
g_1 \rightarrow g_1 + \frac{1}{16\pi^2} & \left(\frac{19}{12} g_1^3 - \frac{1}{6} \sum_{\mathbf{p}} g_1^3 \text{LF}_{2,0} [m_{\tilde{d}}^{\mathbf{p}}] + \frac{1}{9} \sum_{\mathbf{p}} g_1^3 \text{LF}_{3,-1} [m_{\tilde{d}}^{\mathbf{p}}] - \right. \\
& \frac{1}{2} \sum_{\mathbf{p}} g_1^3 \text{LF}_{2,0} [m_{\tilde{e}}^{\mathbf{p}}] + \frac{1}{3} \sum_{\mathbf{p}} g_1^3 \text{LF}_{3,-1} [m_{\tilde{e}}^{\mathbf{p}}] - \frac{1}{4} \sum_{\mathbf{p}} g_1^3 \text{LF}_{2,0} [m_{\tilde{l}}^{\mathbf{p}}] + \frac{1}{6} \sum_{\mathbf{p}} g_1^3 \text{LF}_{3,-1} [m_{\tilde{l}}^{\mathbf{p}}] - \\
& \frac{1}{12} \sum_{\mathbf{p}} g_1^3 \text{LF}_{2,0} [m_{\tilde{q}}^{\mathbf{p}}] + \frac{1}{18} \sum_{\mathbf{p}} g_1^3 \text{LF}_{3,-1} [m_{\tilde{q}}^{\mathbf{p}}] - \frac{2}{3} \sum_{\mathbf{p}} g_1^3 \text{LF}_{2,0} [m_{\tilde{u}}^{\mathbf{p}}] + \\
& \left. \frac{4}{9} \sum_{\mathbf{p}} g_1^3 \text{LF}_{3,-1} [m_{\tilde{u}}^{\mathbf{p}}] - \frac{1}{4} g_1^3 \text{LF}_{2,0} [m_{\oplus}] + \frac{1}{6} g_1^3 \text{LF}_{3,-1} [m_{\oplus}] - \frac{1}{3} g_1^3 \text{LF}_{3,-1} [\tilde{\mu}] \right)
\end{aligned}$$